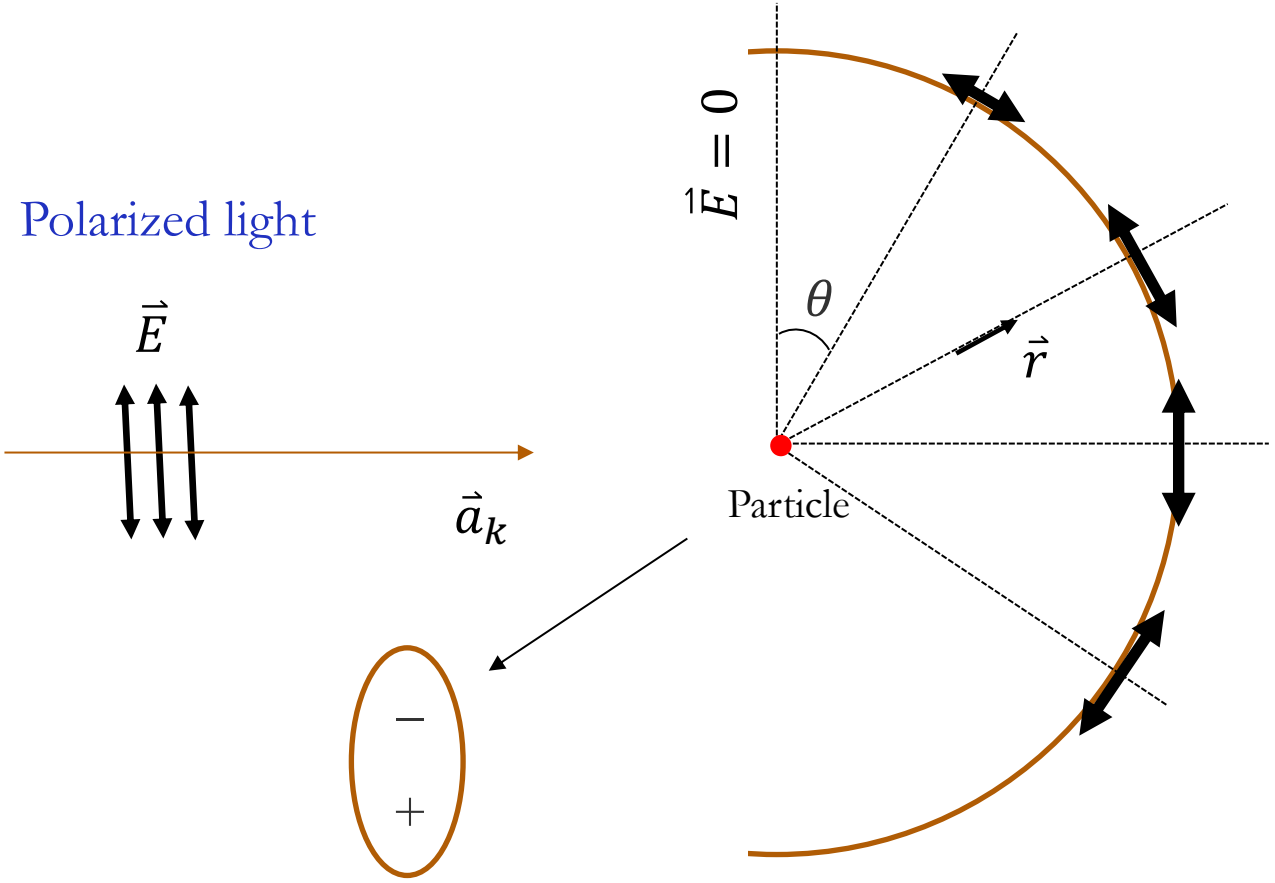
Electromagnetic Wave Optics

Part II: Light Behavior and Properties

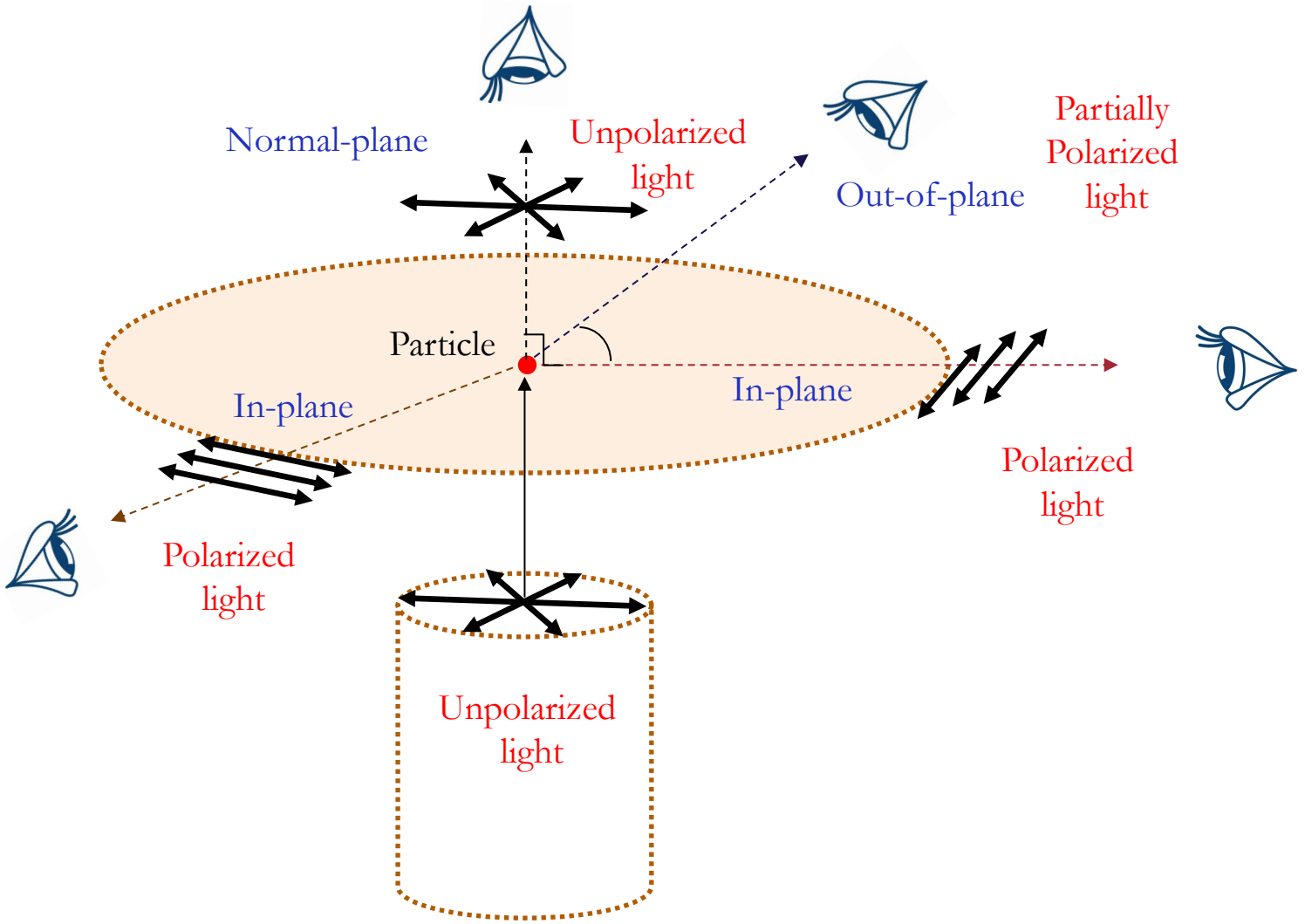
The following subjects will be covered

- Reflection and Transmission of Light.
- Light Polarization.
- Light-matter Interaction.
- Scattering of Light.

EM Wave Scattering



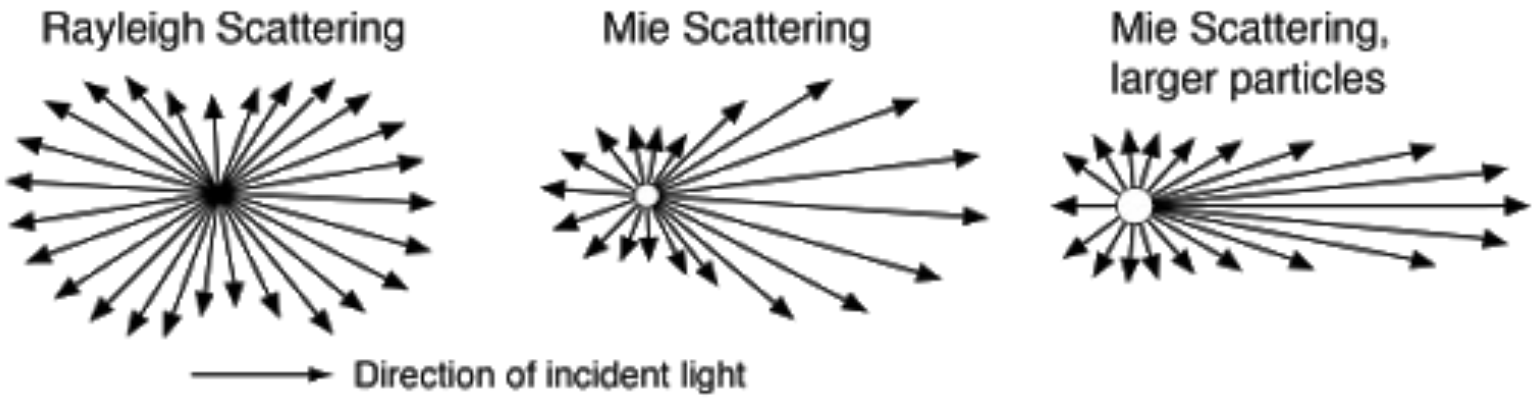
EM Wave Scattering



EM Wave Scattering

Scattering Forms

The wavelength-particle size relation determines the expected form of light scattering



EM Wave Scattering

Rayleigh Scattering

Scattering is a very strong function of wavelength.

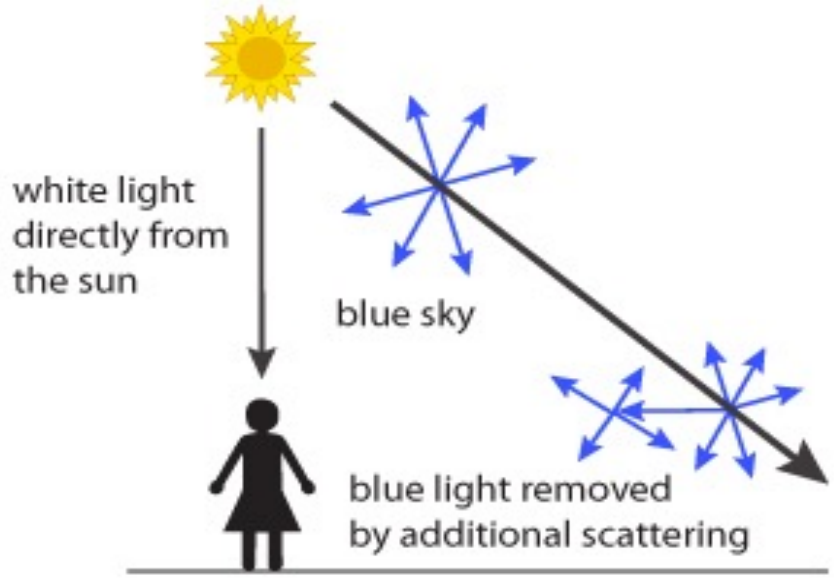
$$I \propto \frac{1}{\lambda^4}$$

Dominant for particles of size $< \frac{1}{10}$ the incident light wavelngth.

Explains the Blue sky!

Air particle and gas molecules have size ~ 0.1 μm.

Blue light wavelength (400 nm) is about 10 times scattered than red light (wavelength around 700 nm).



EM Wave Scattering

Mie Scattering

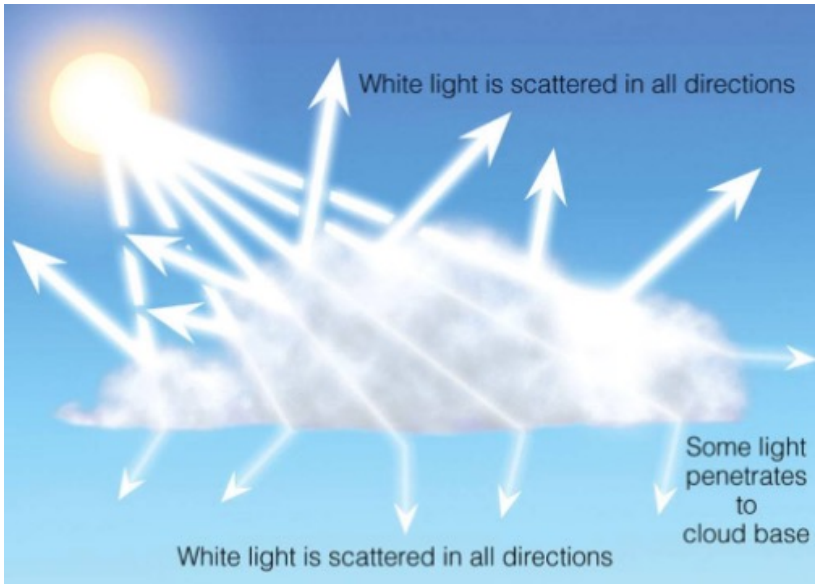
Scattering is not a very strong function of wavelength.

Dominant for particles of size > the incident light wavelngth.

Explains the white clouds!

Water/fog particles have size ~ 10 μm.

All wavelengths scatter nearly the same.



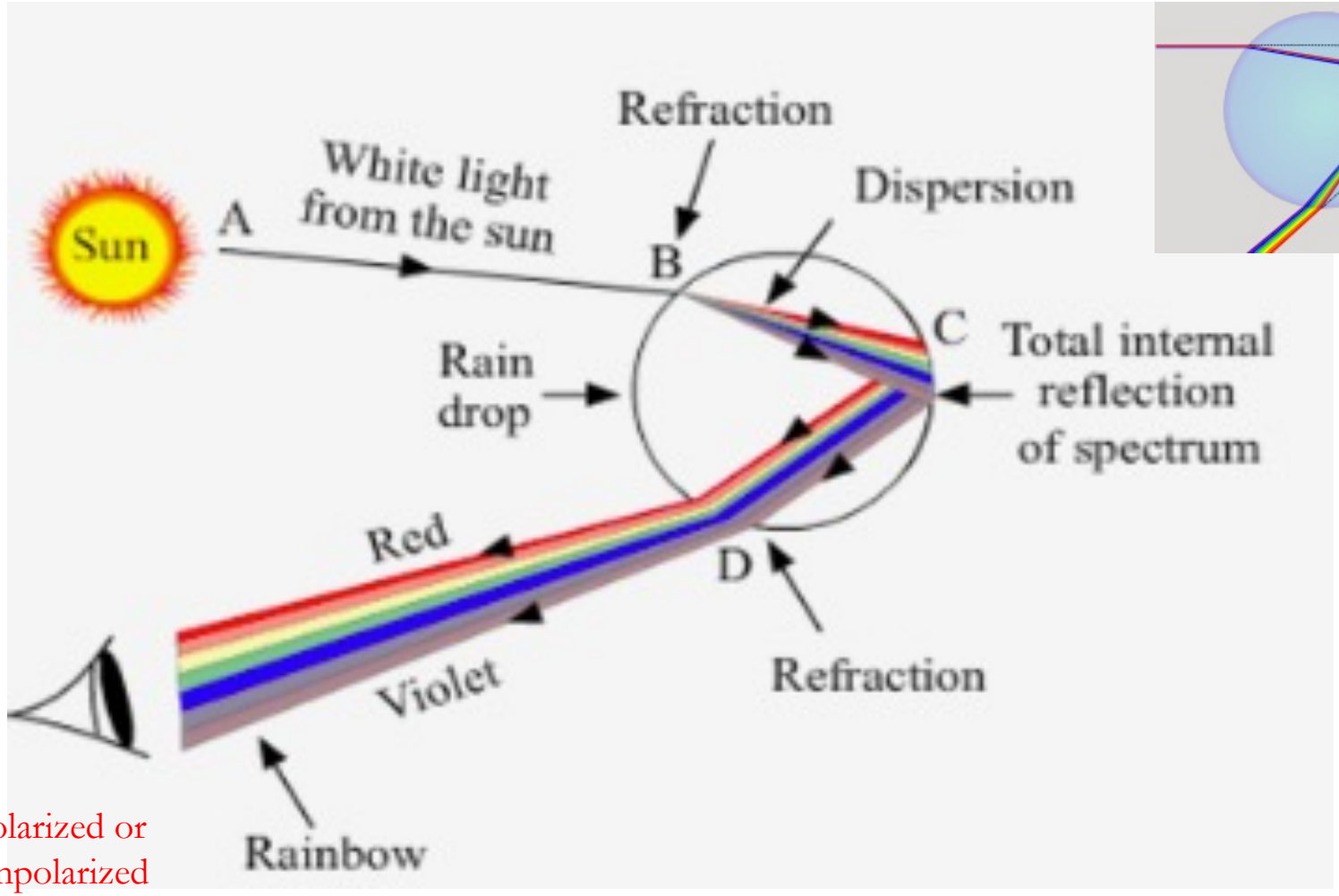
EM Wave Scattering

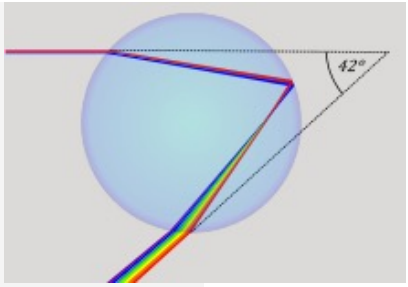


EM Wave Scattering



EM Wave Scattering





Polarized or unpolarized light?

Quantum Optics

The following subjects will be covered

- The Birth of QM
- The Photoelectric Effect
- Light Wave-Particle Duality.
- Light as Photons.
- Examples

The Birth of Quantum Mechanics

The process of generating light from heat was very well "known" BUT
the actual physics was not "understood"
The relation between Temperature and Light intensity and wavelength was not developed.



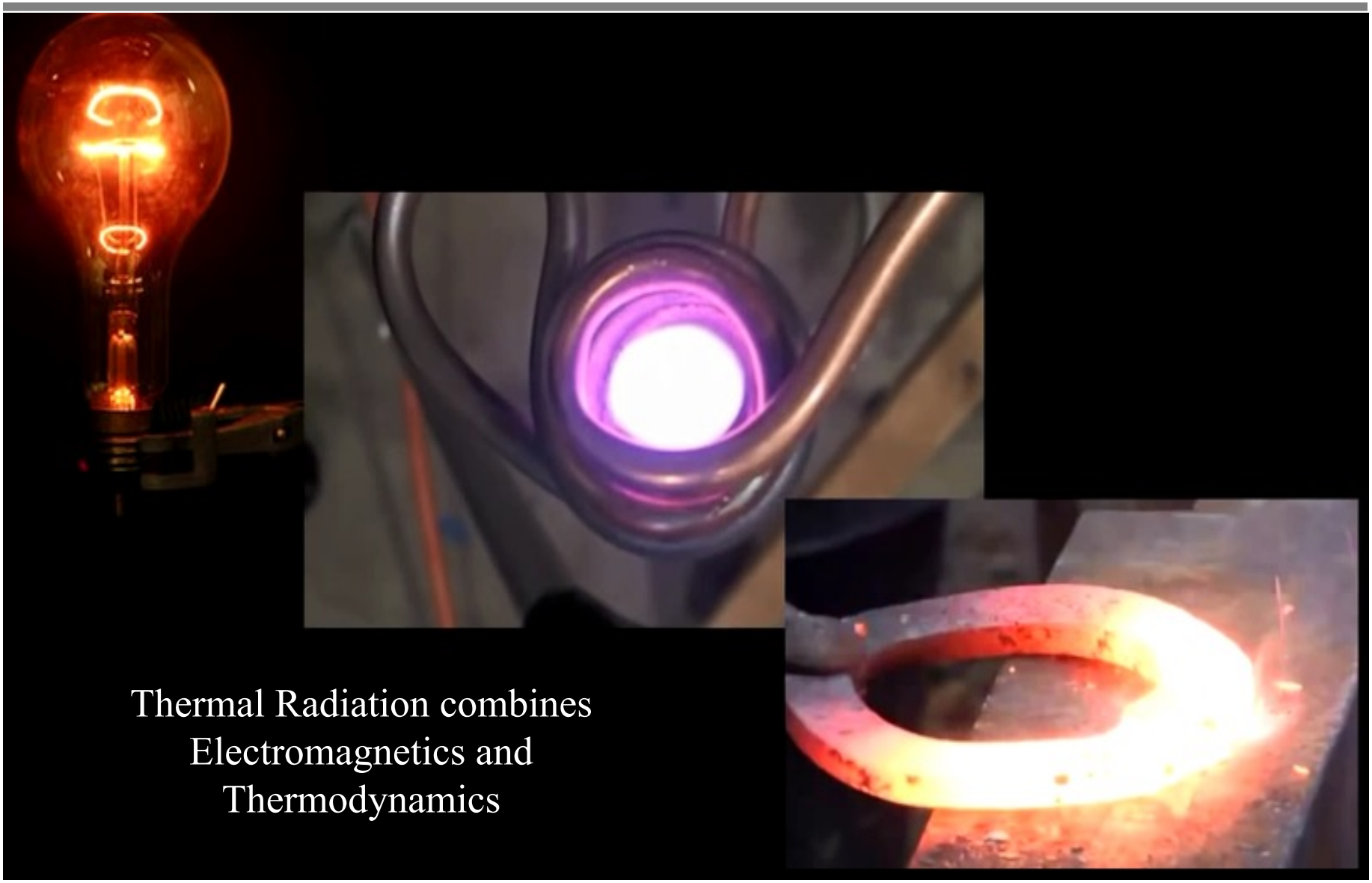
Planck began studying physics in 1874 after being advised against this by a physics professor on the grounds that "*in this field, almost everything is already discovered, and all that remains is to fill a few holes.*"

His doctoral thesis concerned thermodynamics.

In 1894 Planck was commissioned to research how to obtain the most output from a light bulb for a given input of electrical power.

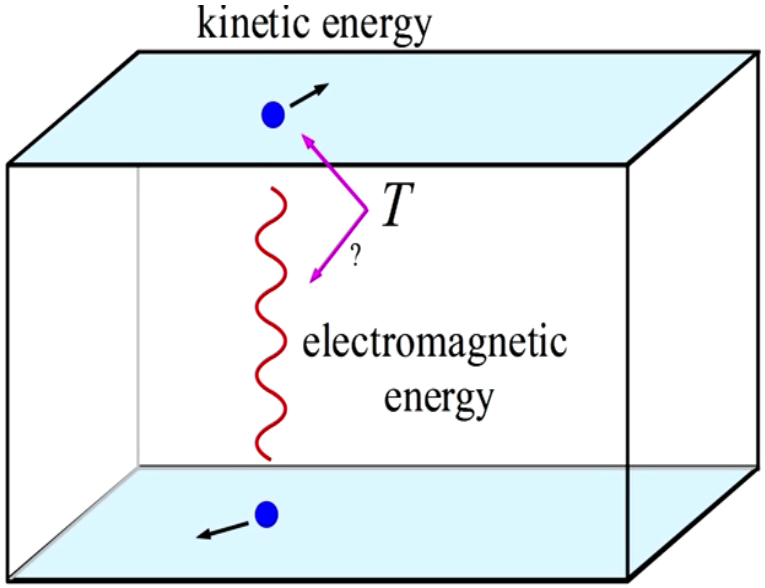


The Birth of Quantum Mechanics



Thermal Radiation combines
Electromagnetics and
Thermodynamics

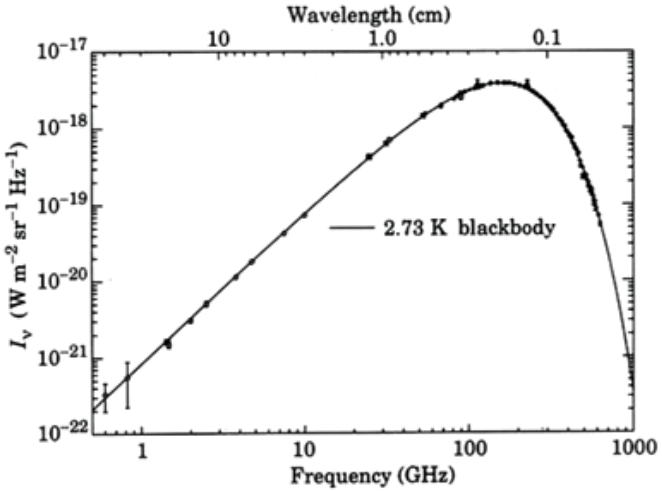
The Birth of Quantum Mechanics



Rayleigh-Jeans law

$$u(\nu) = A\,\nu^2 kT$$

(straight-line on a log-log plot)



Each mode sho Classical physics $u(\nu) = A\,\nu^2 kT$ nergy kT

In thermodynan more modes at $\langle E \rangle_{\text{mode}} = kT$ diation to
increase with te high frequencies

The Birth of Quantum Mechanics

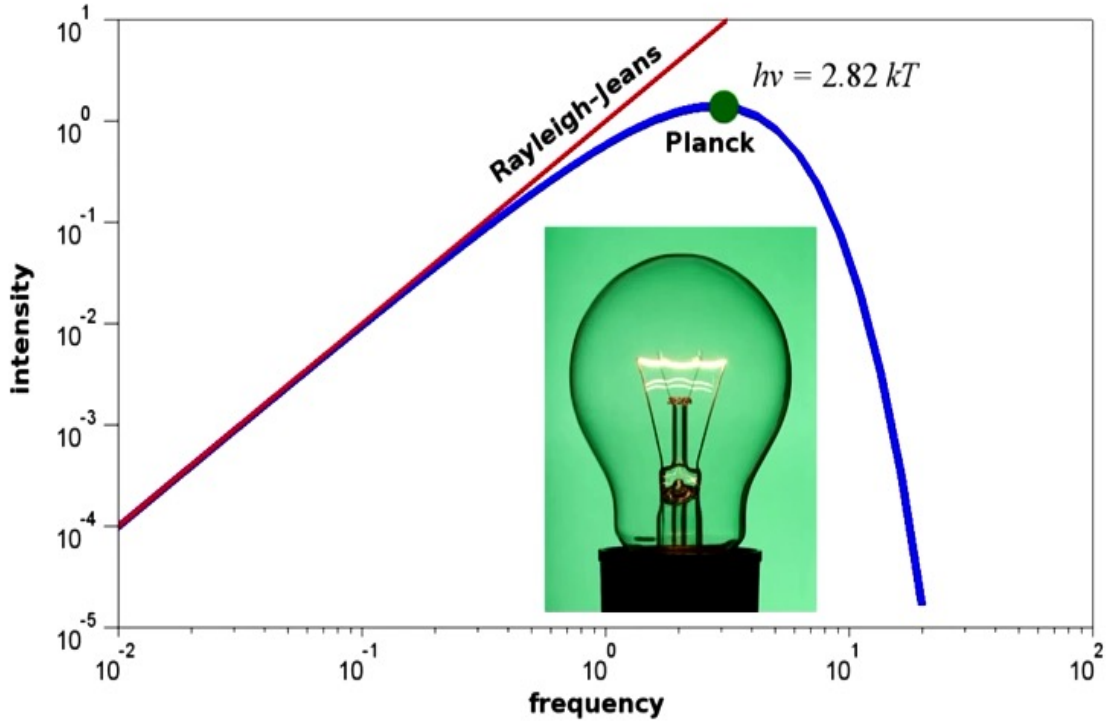
for finite h

$$u(\nu) = A\,\nu^2 \frac{h\nu}{e^{h\nu/kT} - 1}$$

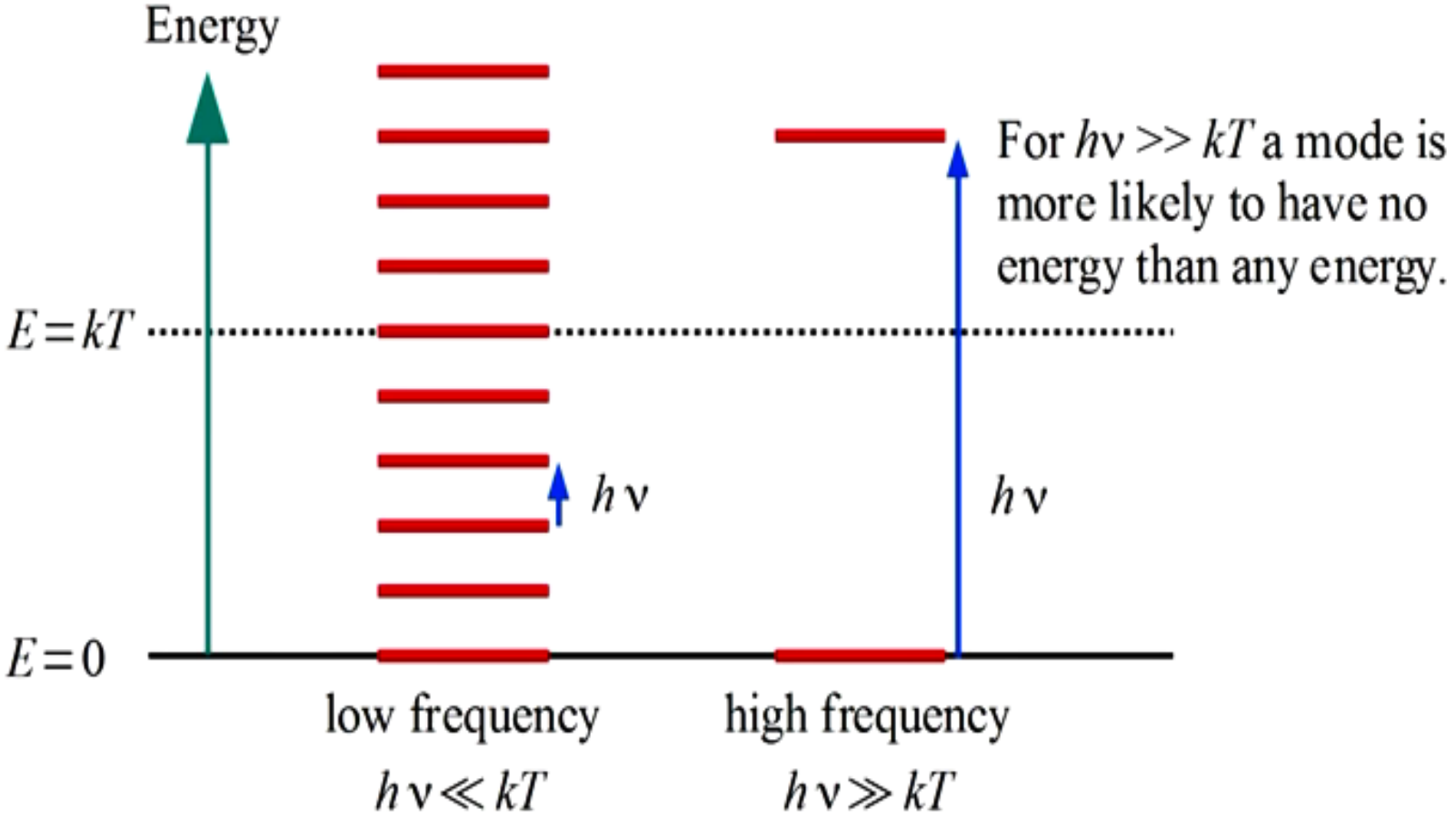
Planck's law (1900)

$$E = n h\nu$$

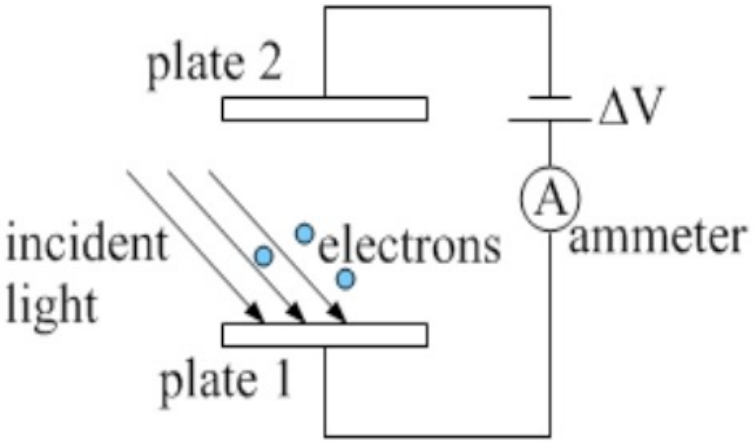
$$h = 6.626 \cdot 10^{-34}\,\text{Js}$$

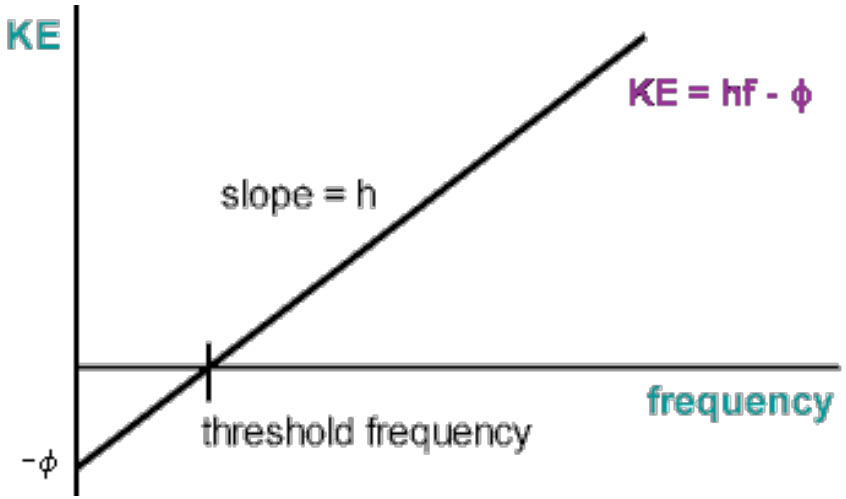


The Birth of Quantum Mechanics



The Photo-electric Effect

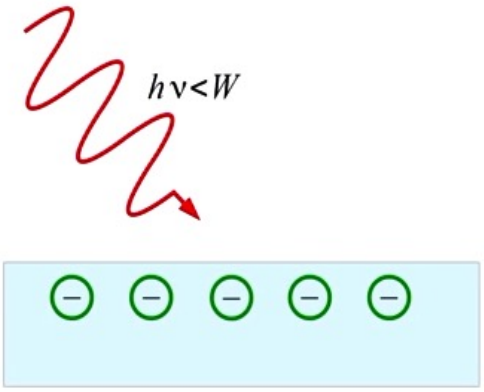


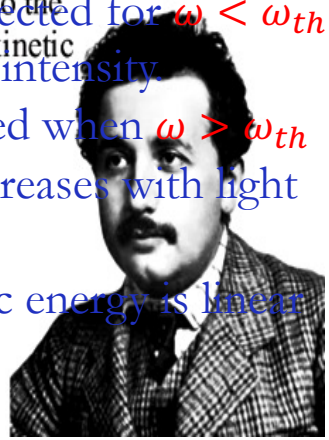


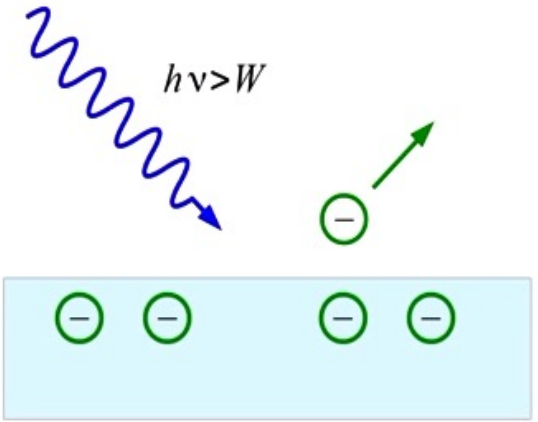
Einstein 1905: If W is the energy needed to remove an electron, only photons with $h\nu > W$ will eject electrons. The number of ejected electrons is proportional to the number of photons. The kinetic energy is $h\nu - W$.

Observations

- No current detected for $\omega < \omega_{th}$ even with high intensity.
- Current detected when $\omega > \omega_{th}$ and current increases with light intensity.
- Electron kinetic energy is linear with ω.

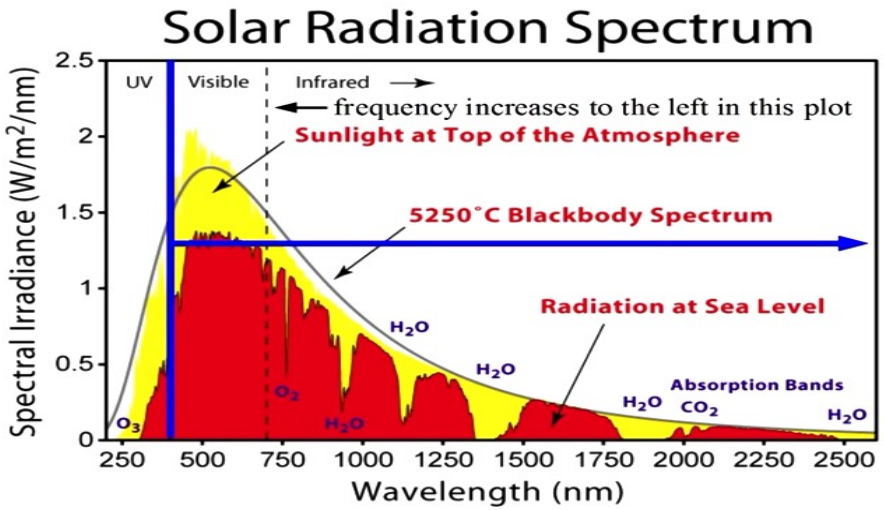


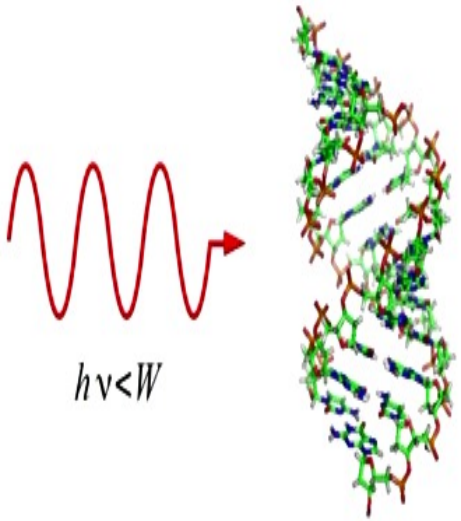




The Birth of Quantum Mechanics



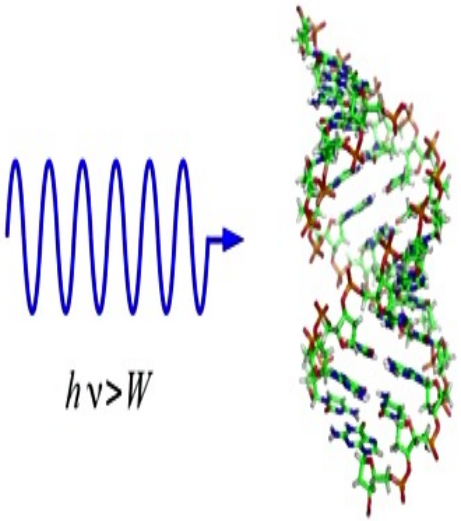




$h\nu < W$

no chemical reaction

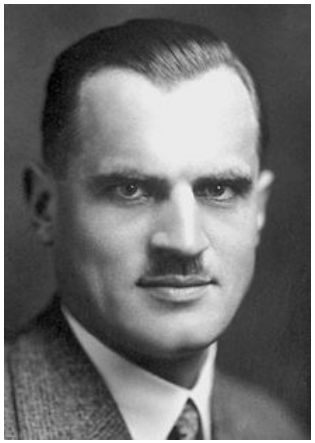
damaging reaction
requires energy W

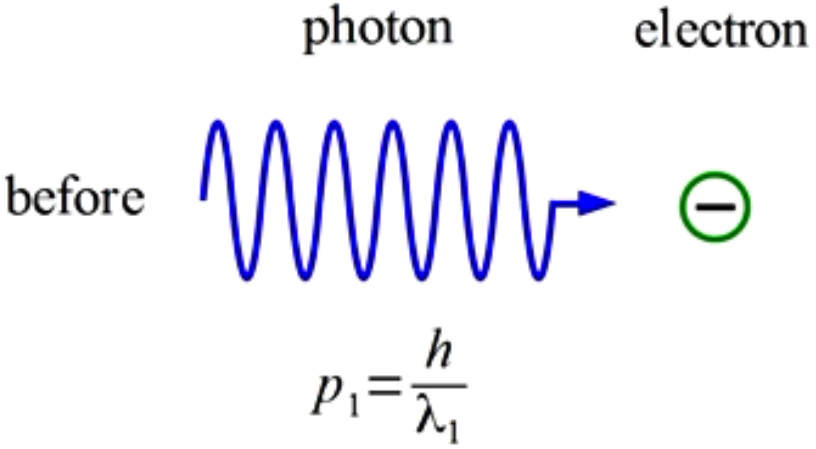


$h\nu > W$

chemical reaction

The Compton Scattering





$$p_1 = \frac{h}{\lambda_1}$$

The Light Quanta: The Photon

Based on these observations, the photon concept emerges

What is a photon?

A light wave with frequency ω can be thought of as composed of a stream of photons (energy particles) that have the same wave speed (c) and travel in the same wave direction.

Photon Energy $E = h f = \hbar \omega = hc/\lambda$

Photon Momentum $p = \frac{\hbar \omega}{v} = \frac{h}{\lambda}$

Photon mass = 0

Photon spin = 1

Photon Stream

Average Photon Flux density in a light wave $\varphi = \frac{I}{\hbar \omega}$ $\frac{photons}{s \cdot m^2}$

Average Photon Flux $\int \varphi \, dA$ $\frac{photons}{s}$

Star light 10^6

Moon light 10^8

Sun light 10^{14}

Laser light 10^{22}

$\frac{photons}{s \cdot cm^2}$

Plank's constant
$h = 6.626 \times 10^{-34}\ Js$

Reduced Plank's constant ($h/2\pi$)
$\hbar = 1.055 \times 10^{-34}\ Js$

The Light Quanta: The Photon

Example. What is the energy in a radiation mode at $\lambda = 1\ \mu m$, in free space?

$\lambda = 1\ \mu m$ ⟶ $\omega = 2\pi\ c/\lambda$

Infrared

$\omega = 2\pi \times 3 \times 10^{14}$ rad/s

Photon Energy $E = \hbar\ \omega = 1.99 \times 10^{-19}\ J$

$$E = \frac{1.99 \times 10^{-19}}{q} = 1.24\ eV$$

Example. What is the photon flux in a 0.2-μm light, if the total power over some area is 1 nW?

$\lambda = 0.2\ \mu m$ ⟶ $\hbar\ \omega = 10 \times 10^{-19}\ J$

UV-C

$$\text{Photon flux} = \frac{1 \times 10^{-9}}{10 \times 10^{-19}} = 1 \times 10^{9} \quad \frac{photons}{s}$$

Example

Ultraviolet light with a wavelength of 240 nm illuminates plate 1 in the figure. Consequently, electrons are emitted and driven across the gap to plate 2 and hence current is read on the ammeter. The battery voltage is gradually increased until the current drops to zero. The battery voltage at this point is 1.2 V.

Find,

1) The energy of the illuminating light.

$$E = 1.24/0.24 = 5.17\ eV$$

2) Maximum kinetic energy of the electron beam.

$$KE = 1.2\ eV$$

3) The work function of the metal.

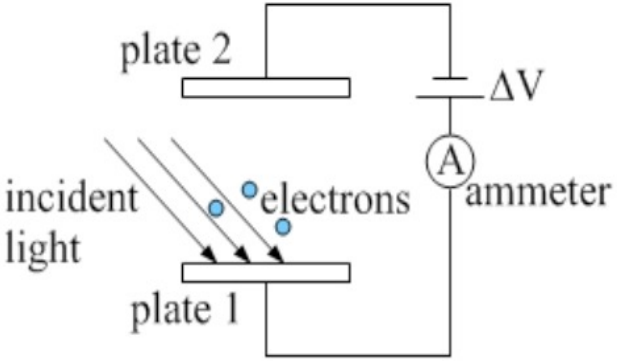
$$W = 5.17 - 1.2 = 3.93\ eV$$

4) The metal used for plate 1.

$$Al$$

5) The maximum wavelength that causes current.

$$\lambda_{max} = ch/\omega = 330\ nm$$



Metal	Work Function, ϕ, (in eV)
Na	2.28
Al	4.08
Cu	4.70
Zn	4.31
Ag	4.73
Pt	6.35
Pb	4.14
Fe	4.50

Radiation Pressure

Light pushes on things!

The classical (wave) concept of wave propagated power can be blended with the quantum mechanical concept to show a "classical" radiation pressure, as follows.

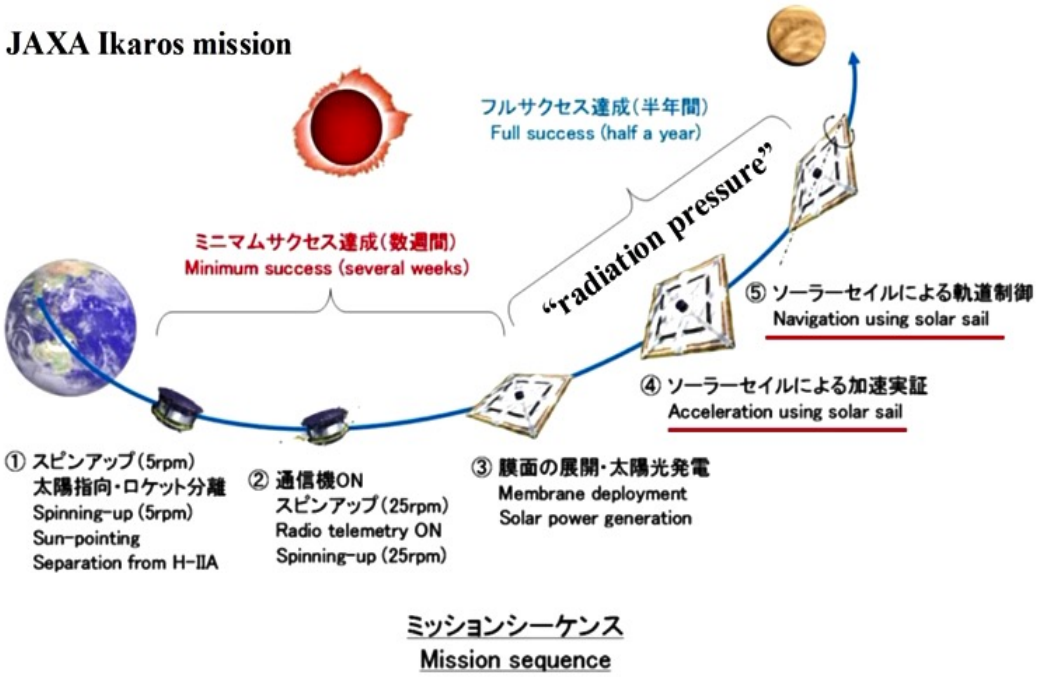
$$\frac{Power\ density\ (S_{ave})}{velocity\ (v)} = \frac{Power}{area}\frac{1}{v}$$

$$= \frac{Energy}{v}\frac{1}{area \cdot time}$$

$$= \frac{Momentum}{time}\frac{1}{area}$$

$$= \frac{Force}{area}$$

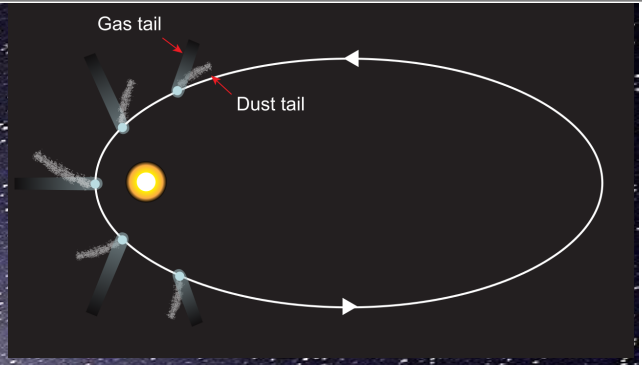
$$= Pressure$$



The solar sail

The Hale-Bopp Comet





Gas tail

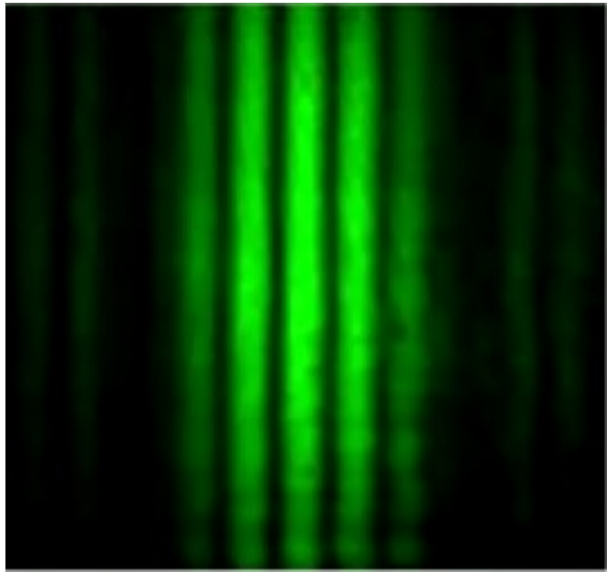
Dust tail



Wave-particle Duality



The Quantum World



Q & A

Send me an email if you have any questions.